

SBT-DF204 Case Study 1 — Investigating Harassment Email Traffic With Wireshark

International Cybersecurity and Digital Forensics Academy (ICDFA) · SBT-DF204
Computer Forensics Case Studies

Student	Ibrahim Diseh Garba
Registration No.	2025/FWSD/11521
Instructor	Aminu Idris, AMCPN · Batch BATCH-B2025 · L1/S2
Date of submission	5 October 2026

1. Case Reference and Student Identification

Field	Detail
Full name	Ibrahim Diseh Garba
Registration No.	2025/FWSD/11521
Programme	Fellowship in Web Application Security & Digital Forensics
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Investigating Harassment Email Traffic With Wireshark
Instructor	Aminu Idris, AMCPN
Scenario	Nitroba University Harassment Scenario (Digital Corpora)
Tools	Kali Linux 2025.3 (VMware), Wireshark, TShark

Declaration. This report is my own work. All filters, packet inspection and conclusions were produced on a hashed working copy of the supplied capture. I did not contact any host, account or person referenced in the evidence, and I make no allegation about any real person outside the training scenario. Unrelated personal data and authentication tokens were redacted from screenshots.

2. Executive Conclusion

The capture shows that the device with internal IP address **192.168.15.4** and Ethernet source MAC **00:17:f2:e2:c0:ce** sent an HTTP POST to **www.willselfdestruct.com (69.25.94.22)** in frame **83601** at **2008-07-22 02:04:24.311700 (EDT, UTC-4)**. The form data is addressed to **lilytuckrige@yahoo.com**, with the subject **"you can't find us"** and message content beginning **"and you can't hide from us."** The server replied with a 302 redirect to **/secure/success**, and the client requested that page in frame 83614.

Traffic from the same MAC address earlier in the capture contains a Gmail chat cookie — **gmail=chat=jcoachj@gmail.com/475090** — observed in frames **79023, 79030, 79032, 79037, 79044, 79049, and 79053**, beginning at **2008-07-22 02:01:03.023427 EDT**. This links the device to the Gmail address **jcoachj@gmail.com**, which is a strong pattern-match to **Johnny Coach** on the Chemistry 109 roster (initial "j" + surname "coach" + trailing initial "j").

Confidence: High (95%) that the device at MAC **00:17:f2:e2:c0:ce** submitted the POST. **Moderate-to-High (75-85%)** that the operator was the person named by the cookie, because the network is open and shared, devices can be shared, and cookies and MAC addresses do not prove who was at the keyboard.

3. Evidence Acquisition and Integrity

Field	Value
Original file name	nitroba_original.pcap
Source	https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Scenario page	digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario
File size	56,180,821 bytes
Download date/time	2026-10-05 06:47:16 EDT

SHA-256 (calculated, original)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA-256 (calculated, working copy)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Result	The two calculated values are identical

The original remained untouched in `evidence/`. All analysis and screenshots used `working/nitroba_working.pcap`.

Evidence ID: E01 · Figure A0

4. Method

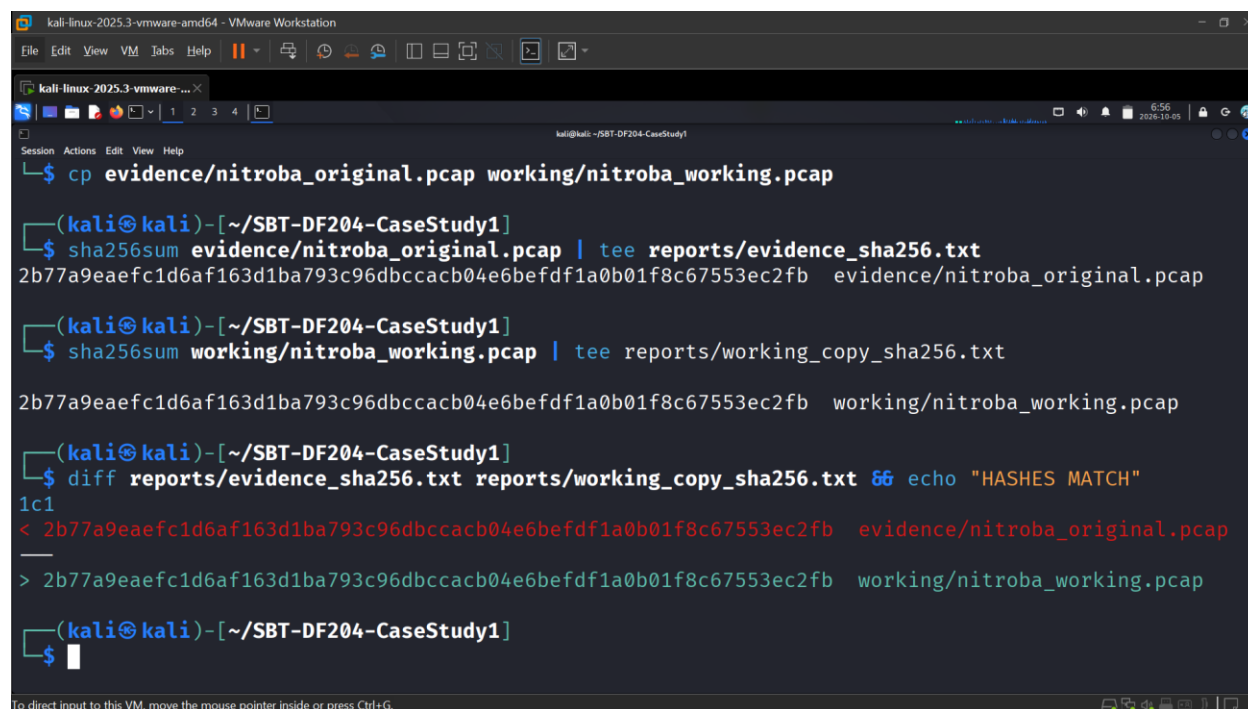
Step	Action	Filter / command
1	Verify integrity	<code>sha256sum</code> on original and working copy
2	Locate <code>willselfdestruct</code> traffic	<code>http.host contains "willselfdestruct"</code>
3	Isolate the POST	<code>http.request.method == "POST" && http.host contains "willselfdestruct"</code>
4	Extract form data	<code>frame.number == 83601</code> with <code>-e urlencoded-form.key -e urlencoded-form.value</code>
5	Reconstruct the dialogue	<code>tshark -q -z follow,tcp,ascii,1707</code>
6	Identify the device	<code>frame.number == 83601</code> with <code>-e eth.src</code>
7	Find identity evidence	<code>eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "gmail"</code>
8	Build timeline	<code>-e frame.time</code> across the filtered frames

The first filter returned ten frames, the third isolated a single POST, and the stream number (**1707**) came from the `tcp.stream` field of that POST.

5. Findings

Q1. What file did you acquire, and how did you preserve its integrity?

See Section 3. The file is `nitroba.pcap`, 56,180,821 bytes, with SHA-256 `2b77a9ea...53ec2fb` recorded for both the original and the working copy.



```
kali@kali: ~/SBT-DF204-CaseStudy1
$ cp evidence/nitroba_original.pcap working/nitroba_working.pcap

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ sha256sum evidence/nitroba_original.pcap | tee reports/evidence_sha256.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba_original.pcap

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ sha256sum working/nitroba_working.pcap | tee reports/working_copy_sha256.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ diff reports/evidence_sha256.txt reports/working_copy_sha256.txt 66 echo "HASHES MATCH"
1c1
< 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba_original.pcap
> 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$
```

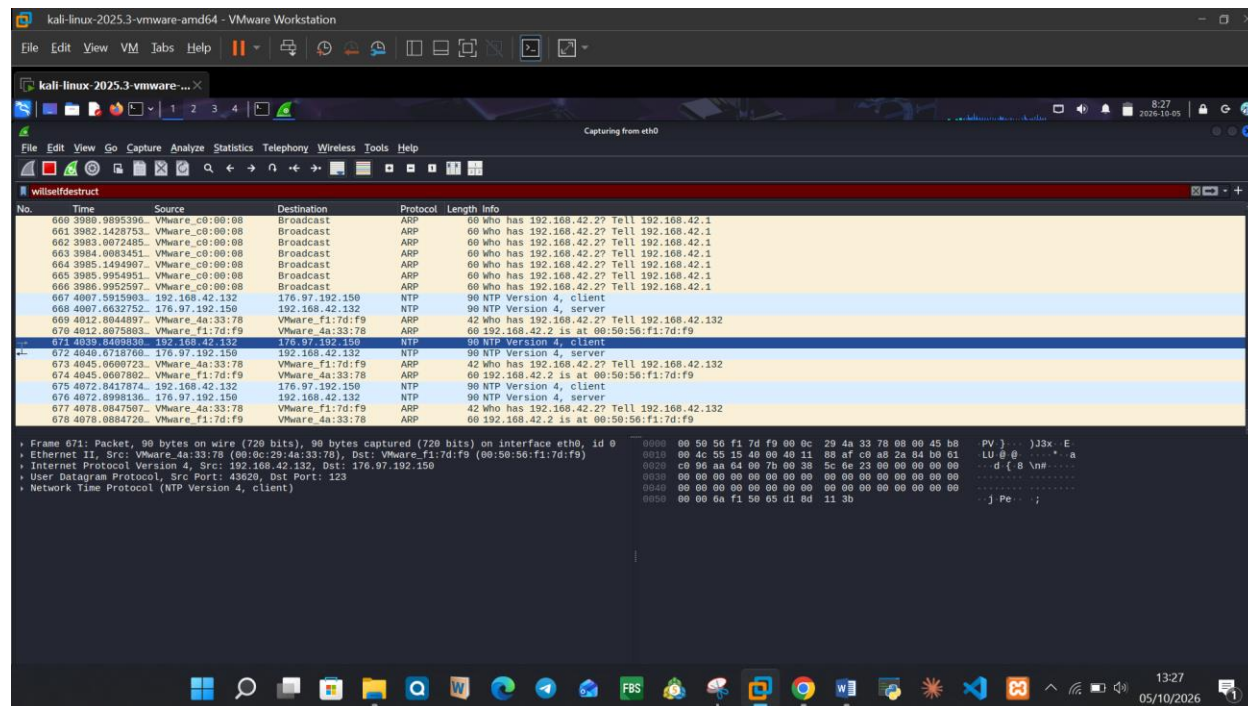
Evidence ID: E01 · Figure: A0

Q2. Which client system contacted the web service?

Property	Value	Evidence
Client IP	192.168.15.4	ip.src in frames 82936 and 83601
Service IP	69.25.94.22	ip.dst in the same frames
Service host	www.willselfdestruct.com	http.host
Client port / stream	36044 / tcp.stream 1707	Follow TCP Stream header, stream 1707

Filter: `http.host` contains `"willselfdestruct"`. The client first fetched the submission form (GET `/secure/submit`, frame 82936, `2008-07-22 02:03:43.825871`) and later submitted it

(POST, frame 83601, 02:04:24.311700). Ten frames matched, all between the same two addresses.



Evidence ID: E02 · Figure: A1

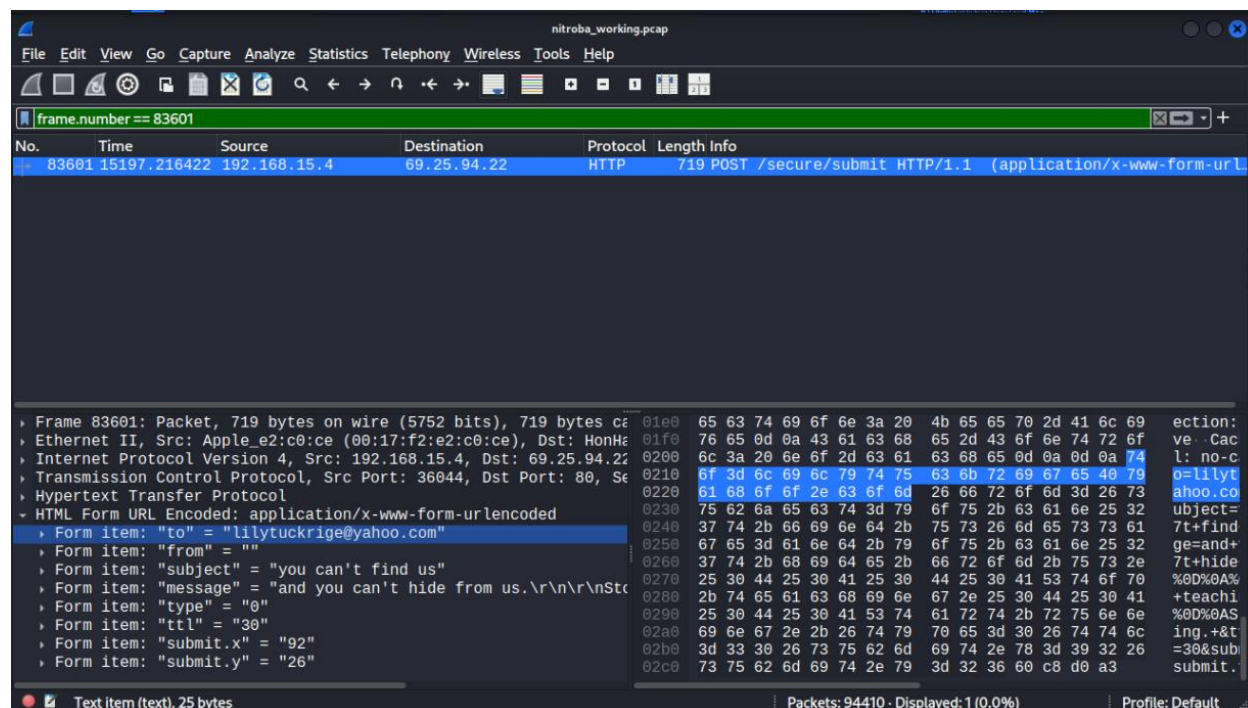
Q3. What evidence links the client to the harassment message?

Frame 83601 (POST /secure/submit, tcp.stream 1707, Content-Length 188) carries URL-encoded form data:

Field	Value
to	lilytuckrige@yahoo.com
from	(empty)
subject	you can't find us
message	and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running.
type / ttl	0 / 30
submit.x / submit.y	92 / 26

Filter: frame.number == 83601. Follow TCP Stream (stream 1707) shows the same body in the request: Host www.willselfdestruct.com, Referer http://www.willselfdestruct.com/secure/submit, and User-Agent Mozilla/4.0

(compatible; MSIE 6.0; Windows NT 5.1; SV1). The server replied HTTP/1.1 302 Moved Temporarily with Location: <http://www.willselfdestruct.com/secure/success>. The recipient address matches the complainant's reported Yahoo account. The capture contains the submitted text, but it cannot show that the email was delivered or read.



Evidence ID: E03 · Figures: A2, A3

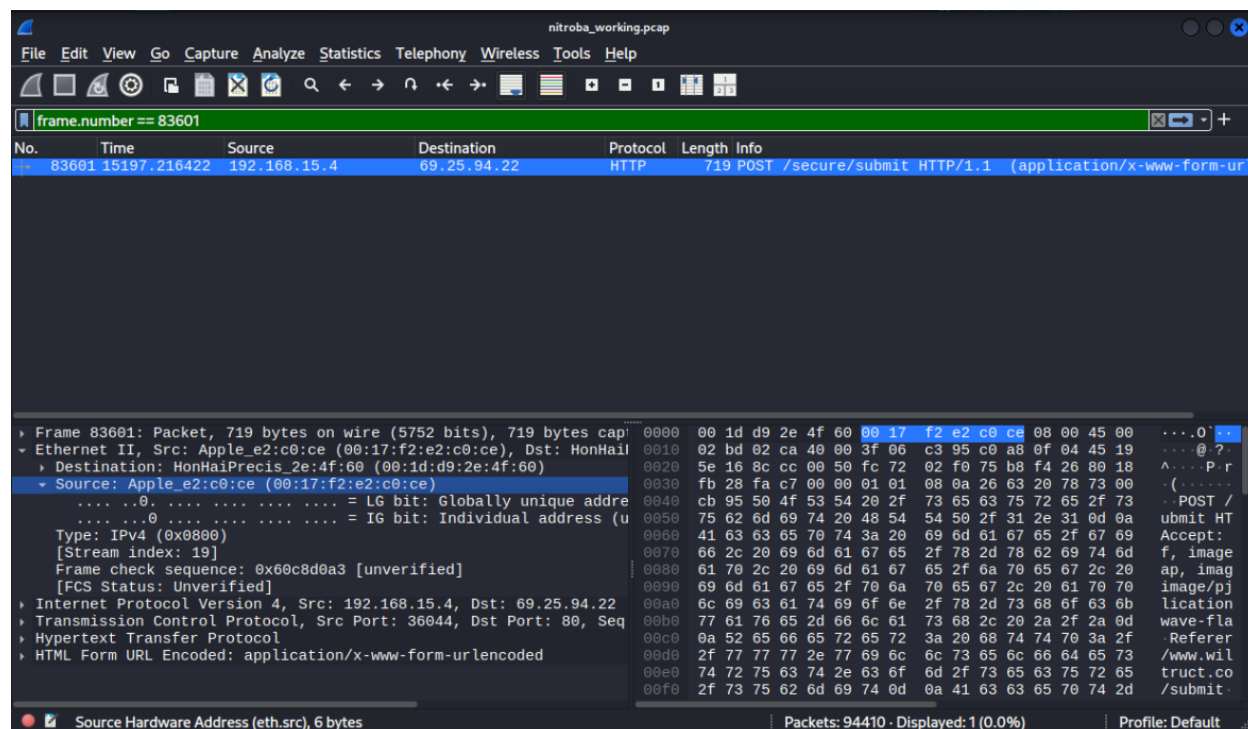
Q4. Which device made the relevant request?

Property	Value	Evidence
Source MAC	00:17:f2:e2:c0:ce	eth.src of frame 83601
OUI	00:17:f2 (registered to Apple, Inc.)	IEEE OUI registry

Why the IP alone proves nothing. 192.168.15.4 is a private (RFC 1918) address behind an open, unpassworded wireless router, so any device in range could obtain one. Outward traffic from all those devices shares the residence's public address. A private IP may also be reassigned between devices. The IP identifies a network position at a moment in time, not a person.

The MAC narrows the traffic to one network interface, but MAC addresses can be changed. One further observation: the User-Agent reports Windows NT 5.1 while the OUI

is Apple's. That is not contradictory (Apple hardware can run Windows via Boot Camp or a virtual machine), but this capture cannot resolve it.



Evidence ID: E04 · Figure: A4

Q5. Who is associated with the device?

Traffic carrying the same MAC was searched for Gmail cookies:

```
bash
tshark -r working/nitroba_working.pcap \
  -Y 'eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "gmail"' \
  -T fields -e frame.number -e frame.time -e http.host -e http.cookie
```

Property	Value
Frame(s)	79023, 79030, 79032, 79037, 79044, 79049, 79053
First observed	2008-07-22 02:01:03.023427000-0400 (frame 79023)
Cookie pair	gmail=chat=jcoachj@gmail.com/475090
Email address	jcoachj@gmail.com

HTTP host	mail.google.com
-----------	-----------------

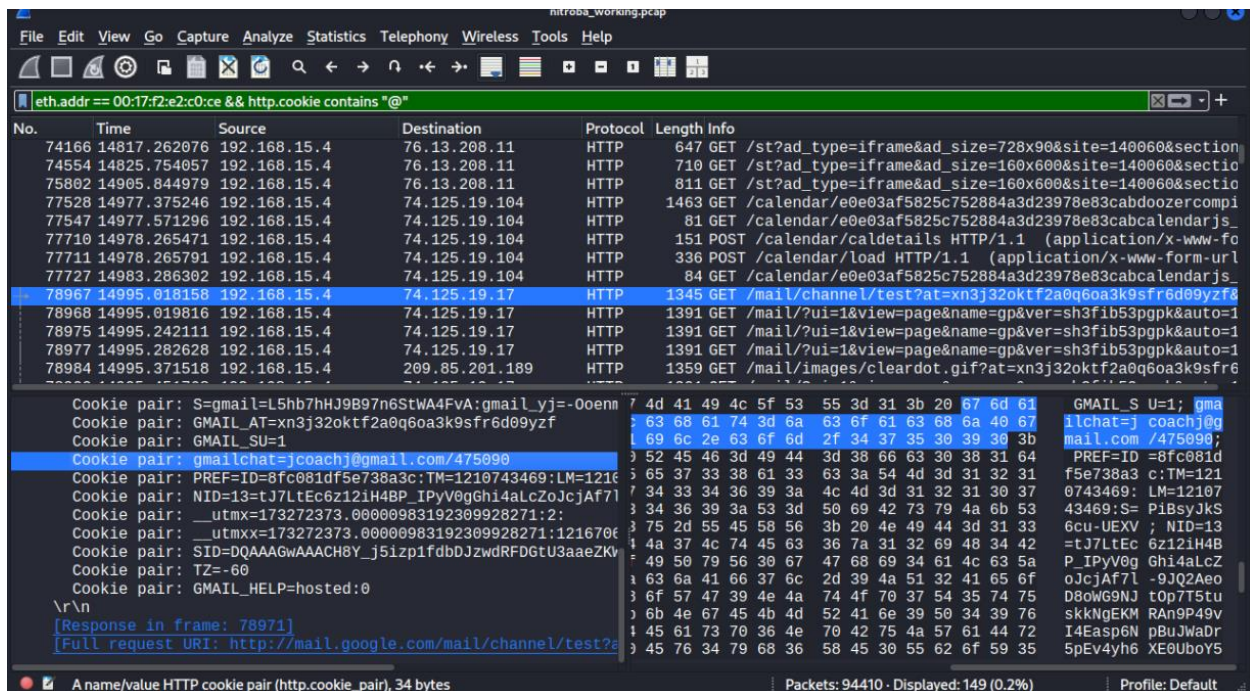
The email address is extracted cleanly with:

```
bash
tshark -r working/nitroba_working.pcap \
  -Y 'eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "gmail"' \
  -T fields -e http.cookie \
  | grep -oE '[a-z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{2,}' | sort -u
```

Output: jcoachj@gmail.com

Note on filter refinement. An earlier broad filter — `http.cookie contains "@"` — matched several unrelated frames (18823, 19159, 34995, 36105, 37574, 59712) whose `@` characters sit inside random token strings in Amazon cookies (ubid-main, x-main, session-token). Those frames were excluded. Narrowing the filter to Gmail cookies isolated the actual identity evidence.

The Gmail chat cookie indicates that a Gmail account was in use in a browser on this device **three minutes before** the harassment POST. It does not prove who owned or operated that browser.



Evidence ID: E06 · Figure: A6

Q6. Is that person on the Chem 109 roster?

The scenario roster lists: Amy Smith, Burt Greedom, Tuck Gorge, Ava Book, **Johnny Coach**, Jeremy Ledvkin, Nancy Colburne, Tamara Perkins, Esther Pringle, Asar Misrad, Jenny Kant.

Comparison of `jcoachj@gmail.com` **with the roster:**

Gmail local part	Roster match	Pattern
<code>jcoachj</code>	Johnny Coach	First initial j + surname coach + trailing initial j

This is a **pattern match on a username**, not confirmation of account ownership, so it is an inference.

Evidence ID: E07 · Figure: A7

Q7. When did the activity occur?

TShark displays times with offset **-0400 (EDT)**. Conversion: **UTC = EDT + 4 hours**.

Event	Frame	EDT (as displayed)	UTC
Gmail chat cookie observed	79023	2008-07-22 02:01:03.023427	2008-07-22 06:01:03.023427
GET /secure/submit (form)	82936	2008-07-22 02:03:43.825871	2008-07-22 06:03:43.825871
POST /secure/submit (message)	83601	2008-07-22 02:04:24.311700	2008-07-22 06:04:24.311700
GET /secure/success	83614	2008-07-22 02:04:24.564165	2008-07-22 06:04:24.564165
GET /images/bk-message.gif	83654	2008-07-22 02:04:24.902157	2008-07-22 06:04:24.902157

Two timing observations remain unreconciled. First, the server's 302 response carries Date: Tue, 22 Jul 2008 07:24:45 GMT, about 1 h 20 min later than the PCAP-derived UTC time of the POST (06:04:24). This indicates a clock offset on the server or the capture host, and I did not determine which. Second, the brief states the later message arrived on Monday 21 July, while the PCAP timestamps fall on Tuesday 22 July. All times in this report are therefore **PCAP timestamps**, and they should not be treated as proven real-world time.

6. Timeline

Times are PCAP timestamps (EDT as displayed; UTC = EDT + 4 h).

#	EDT	UTC	Frame	Event
1	2008-07-22 02:01:03.023427	06:01:03.023427	79023	Gmail chat cookie observed (jcoachj@gmail.com)
2	02:03:43.825871	06:03:43.825871	82936	GET /secure/submit
3	02:03:44.125949	06:03:44.125949	82985	GET /images/spacer.gif
4	02:03:44.334254	06:03:44.334254	83025	GET /images/sm-logo.gif
5	02:03:44.374583	06:03:44.374583	83037	GET /images/warning-home.gif
6	02:03:44.530352	06:03:44.530352	83072	GET /images/body-bk.gif
7	02:03:44.599079	06:03:44.599079	83087	GET /images/bttn-send.gif
8	02:03:44.852265	06:03:44.852265	83162	GET /images/bridge_small.gif
9	02:04:24.311700	06:04:24.311700	83601	POST /secure/submit (harassment message)
10	02:04:24.564165	06:04:24.564165	83614	GET /secure/success
11	02:04:24.902157	06:04:24.902157	83654	GET /images/bk-message.gif

The Gmail cookie was observed about **3 minutes 21 seconds** before the POST. The form was loaded about **40.5 seconds** before submission.

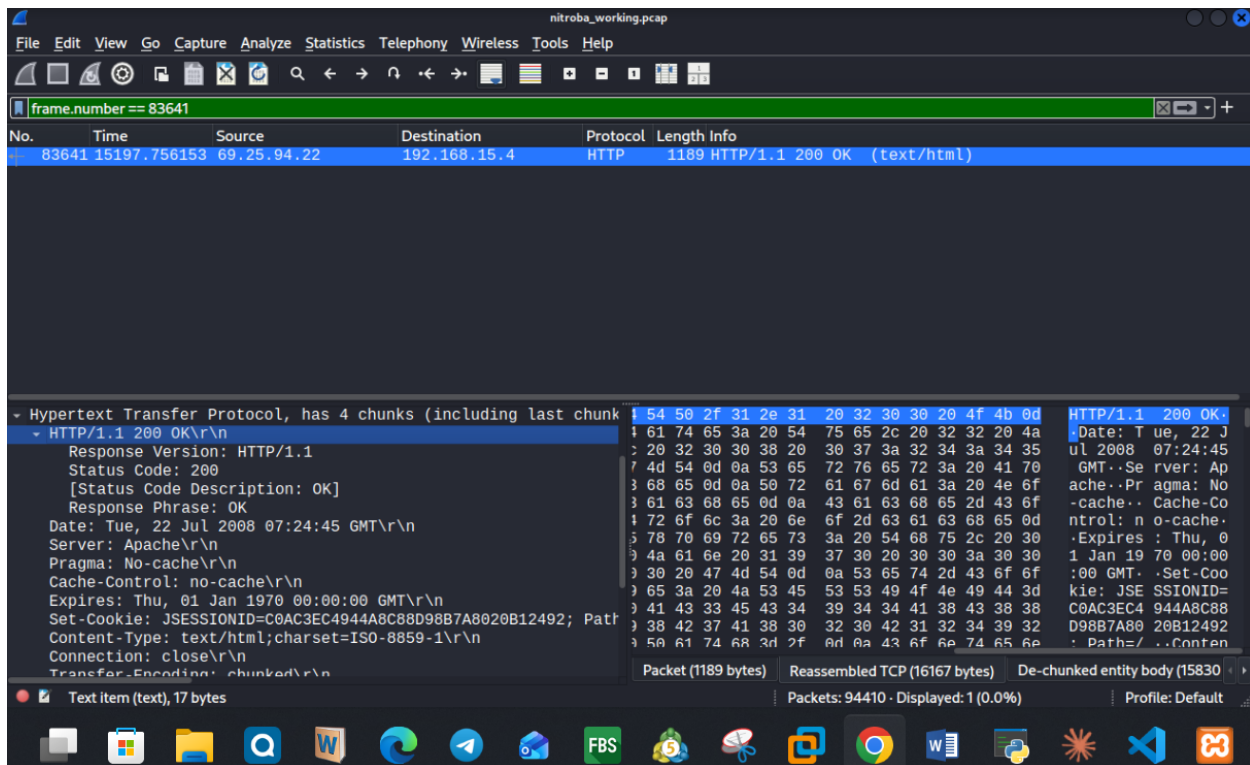


Figure: A8

7. Attribution Assessment

Category	Content
Observed facts	POST in frame 83601 from 192.168.15.4 / 00:17:f2:e2:c0:ce contains the message and recipient; the server returned a 302 to /secure/success; the same MAC carried a Gmail cookie gmail=chat=jcoachj@gmail.com/475090 in frames 79023–79053
Inferences	The Gmail account belongs to the operator; the username maps to roster name Johnny Coach ; that student sent the message
Confidence	High (95%) device; Moderate-to-High (75–85%) individual
Alternatives	Open Wi-Fi use by a third party; shared device or browser; MAC spoofing; stolen or reused session cookie
Would raise confidence	Wireless association logs, device forensics, account records from the provider, witness statements
Would lower confidence	Evidence of MAC spoofing, a shared browser profile, or other users on the same device

8. Recommendations

For investigators: Keep the original untouched, hash every derived file, record the displayed time zone, redact unrelated data, state confidence explicitly, and avoid naming a suspect without a documented evidence link.

For the university: Use WPA2/WPA3 and 802.1X where possible, log wireless associations, restrict unmanaged personal routers in residences, and educate residents about the shared-network attribution problem.

9. Appendices

A. Evidence Log

ID	Packet / item	Finding	Why it matters	Figure
E01	nitroba_original.pcap and working copy	Identical SHA-256 values	Integrity of analysed file	A0
E02	Frame 82936 (and 83601)	Client 192.168.15.4 to 69.25.94.22, <i>willselfdestruct.com</i>	Identifies client and service	A1
E03	Frame 83601, stream 1707	POST containing recipient, subject, message	Links client to message	A2, A3
E04	Frame 83601 eth.src	MAC 00:17:f2:e2:c0:ce	Identifies device	A4
E05	Stream 1707 response and frame 83614	302 redirect to /secure/success, then GET	Shows submission was accepted	A5
E06	Frames 79023, 79030, 79032, 79037, 79044, 79049, 79053	Gmail cookie gmail=chat=jcoachj@gmail.com/475090	Candidate identity	A6
E07	Class roster	jcoachj@gmail.com → Johnny Coach	Chem 109 membership	A7
E08	Frames 79023 to 83654	Timeline	Sequence of events	A8

B. Screenshot Checklist

Each screenshot must show the active filter, the packet list or command, the relevant detail and the time. Redact authentication tokens and unrelated personal data.

Figure	Content	Source
A0	Both SHA-256 values, with the identical-hash check	Terminal
A1	Ten willselfdestruct frames	Wireshark, filter <code>http.host contains "willselfdestruct"</code>
A2	POST isolated, frame 83601, stream 1707	Terminal or Wireshark
A3	Form data and Follow TCP Stream, stream 1707	Terminal or Wireshark
A4	Ethernet II source MAC of frame 83601	Wireshark packet details
A5	302 response and frame 83614	Wireshark
A6	Gmail cookie with <code>gmail=chat=jcoachj@gmail.com/475090</code> highlighted — other cookies cropped out	Wireshark or terminal
A7	Class roster	Scenario file
A8	Timeline output	Terminal

C. Chain-of-Custody Worksheet

Field	Value
Case identifier	SBT-DF204-CaseStudy1-Ibrahim-Diseh-Garba
Examiner / custodian	Ibrahim Diseh Garba (ICDFA trainee)
Evidence files	<code>evidence/nitroba_original.pcap</code> (preserved), <code>working/nitroba_working.pcap</code> (analysis copy)

Source	https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Size	56,180,821 bytes
SHA-256 (original and working)	2b77a9eaeffc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Acquired	2026-10-05 06:47:16 EDT
Handling	Original not analysed directly; no external systems contacted

D. References

1. *Digital Corpora. Nitroba University Harassment Scenario.*
digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario
2. *Wireshark Foundation. Wireshark User Guide.* wireshark.org/docs
3. *TShark manual page.* wireshark.org/docs/man-pages/tshark.html
4. *RFC 6265 (HTTP cookies) · RFC 7231 (HTTP/1.1 semantics) · RFC 1918 (private address space)*
5. *ICDFA. SBT-DF204 Case Study 1 assessment brief.*